

Concept Note for InnoHK Secretariat

Centre for Multi-Modal Physics-Informed Sensing & Inversion (MM-PSI)

Fusing Vision + Sparse Vibration via SciML for Structural Health Monitoring of Large-Scale Infrastructure

Host	The Chinese University of Hong Kong (CUHK), Dept. of Mathematics
Cluster	AIR@InnoHK — AI & Robotics
Lead PI	Prof. ZOU Jun (郑军), CUHK Mathematics
Theme	Multi-Modal SciML for Infrastructure Health Monitoring
Duration	5 Years (Proposed)

1. Background & Motivation

Large-scale composite structures—wind turbine blades and aircraft wings—operate under extreme, unpredictable loads. Their health monitoring faces a dual challenge. Externally, we need low-cost, deployable sensing of surface deformation (e.g., monocular smartphone video). Internally, we need to infer damage (delamination, matrix cracking) from only a handful of accelerometers, under unknown excitation and strongly anisotropic material behavior.

Existing approaches are siloed: computer-vision methods see only the surface and cannot sense internal damage; vibration-based inverse methods require dense sensor arrays or known excitation—assumptions that collapse in the field. There is no unified framework that fuses dense visual observation with sparse vibrational sensing through physics-grounded AI.

2. Core Scientific Question

Can a unified, physics-informed AI framework fuse dense monocular surface-deformation sensing with sparse multi-sensor vibration inversion—using visual deformation fields as boundary-condition priors to constrain and guide internal damage inference—to deliver end-to-end, high-confidence diagnostics for large composite structures?

3. Technical Architecture: Three Streams

Stream A — Visual Deformation Sensing

Input: monocular smartphone video of rotating blade / wing under operation.

Method: three-layer decomposition (rigid motion + aeroelastic modal deformation + high-frequency residual) with physics-data joint optimization; photometric consistency + modal regularization + temporal smoothness; benchmarks against MoGe-2 / Align3R / MegaSaM.

Output: dense 3D deformation field (surface displacement map) at sub-millimetre accuracy.

Stream B — Sparse Vibration Inversion

Input: 6 accelerometers embedded in composite structure.

Method: PINN encoding Navier-Cauchy anisotropic wave equation + GNN encoding sensor-network topology (Fiedler embedding, Chebyshev graph convolution); Explicit Constraint Force Method (ECFM) for ill-posedness resolution; unknown excitation parameterized as latent variables.

Output: damage field (location <5 mm, area <10%) with uncertainty intervals.

Stream C — Multi-Modal Fusion (The Integrative Core)

Boundary-condition coupling: visual deformation field imposes kinematic constraints on the PINN's displacement boundary conditions, drastically narrowing the solution space.

Graph-Laplacian fusion: visual surface strain serves as additional nodes in the sensor graph, enriching topological information.

End-to-end joint optimization: Encode–Process–Decode with semi-implicit Euler; Adam → L-BFGS-B two-stage; Bayesian GNN for UQ.

4. Strategic Programme (4 SPs)

SP	Title	Deliverable
SP1	Visual Deformation Sensing	Monocular 3-layer decomposition; CVPR/ICCV papers
SP2	Sparse Vibration Inversion	PINN+GNN framework; JCP/CMAME/NeurIPS
SP3	Multi-Modal Fusion Inference	Joint optimization engine; open-source toolkit
SP4	UQ & Edge-Cloud Deployment	Bayesian GNN; DT API; edge inference box

5. Global Collaboration

MoU agreements to be executed prior to full submission.

Institution	Expertise	Role
ETH Zurich	SciML, SHM	PINN theory; joint SP2/SP3; PhD exchange
MIT CSAIL	Computer vision, 3D recon.	Monocular depth; SP1; joint papers
MIT (Karniadakis)	PINN originators, SciML	Algorithm co-dev; SP2/SP3
Cambridge Eng.	Composite mechanics	Validation datasets; field trials; SP2
DTU Wind Energy	Wind turbine testing	Full-scale blade experiments; SP1/SP4

6. Impact & Translation

Scientific: establishes the first unified SciML framework for multi-modal (vision + vibration) structural health monitoring—extensible to bridges, pressure vessels, and biomedical implants.

Industrial: sub-USD 1K visual monitoring + 6-sensor vibration inversion—order-of-magnitude cost reduction vs. USD 50K+ multi-camera / dense-sensor systems.

Pathway: InnoHK Y1–2 R&D → RAISE+ Y2–3 spin-off → pilot deployments with CLP Power / China Three Gorges / Airbus MRO.

Talent: 4 PhD + 6 MPhil + 8 postdocs in SciML over 5 years; cross-disciplinary curriculum with CUHK Mathematics and Engineering.

7. Benefit to Hong Kong

Positions CUHK Mathematics as a global leader in multi-modal SciML for infrastructure.

Supports HK’s carbon-neutrality goals via intelligent wind-energy infrastructure.

Creates GBA talent pipeline in AI-driven structural health monitoring.

8. Indicative 5-Year Budget

Personnel (5 yrs)	~35–45M HKD
Equipment (cameras, GPUs, edge devices, blade test rig)	~10–15M
Operations & International Travel	~5–7M
Total (Indicative)	~55–65M HKD

9. Differentiation from Existing AIR@InnoHK Centres

AIR@InnoHK hosts 16 laboratories in generative AI, human–AI interaction, logistics robotics, and medical robotics. None addresses multi-modal physics-informed sensing and inversion for large-scale infrastructure—a capability that underpins the next generation of smart, safe, and sustainable physical infrastructure. MM-PSI, anchored in the Department of Mathematics, provides the algorithmic and theoretical foundation (PINN + GNN + differentiable inference) that complements the perception and decision capabilities of existing centres.

10. Request for Secretariat Feedback

Q1: Is this multi-modal SciML scope—vision + sparse vibration fused through physics-informed AI—appropriately aligned with AIR@InnoHK?

Q2: Should we establish formal affiliation with an existing centre (e.g., CPII) as a sister centre, or submit independently?

Q3: Which KPIs (papers, patents, industry pilots, spin-offs, talent) carry greatest weight in AIR@InnoHK evaluation?

Pending positive feedback, we plan full proposal submission in Q4 2026, accompanied by executed MoUs and industry letters of support.

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